



Exercise 3: CORRELATION MATRIX

Write a python programs to load the dataset and understand the input data

1. Load data, describe the given data and identify missing, outlier data items
2. Find correlation among all attributes
3. Visualize correlation matrix

1. Load data

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv("D:/diabetes.csv")
print(df.describe())
print(df.head(10))
print(df.isnull())
```

2. Find correlation among all attributes

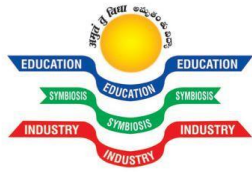
```
import pandas as pd
import matplotlib.pyplot as plt
import scipy.stats

# Making data frame from the csv file
df = pd.read_csv("nba.csv")

# Printing the first 10 rows of the data frame for visualization
print(df[:10])

# To find the correlation among columns # using pearson method
print(df.corr(method='pearson'))

# using „kendall“ method.
print(df.corr(method='kendall'))
```



3. **Visualize correlation matrix**

```
import pandas as pd  
df = pd.read_csv("D:/diabetes.csv")  
print(df. describe())  
print(df.head(10))
```